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MAM1019H

Hand-in 4

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Answers

1. (a) False
   (b) True
   (c) False
   (d) False

2. Some $a \in \mathbb{N} - \{1\}$ is minimal if there is no $x \in \mathbb{N} - \{1\}$ such that $x \neq a$ and $x|a$. Now, this means that $p \in \mathbb{N}$, some prime number, is minimal, because there is no $x \in \mathbb{N} - \{1\}$ such that $x|p$ (otherwise it is not a prime number). Now suppose there is some composite number c which is minimal. Then $\exists p \in \mathbb{N}, p$ is a prime number, and $p \neq a$ and $p|a$ so a is not minimal, so the set of all minimal elements is the set of all prime numbers.
   Now some $a \in \mathbb{N} - \{1\}$ is maximal if there is no $x \in \mathbb{N} - \{1\}$ with $x \neq a$ and $a|x$. Now this means that 0 is maximal, because there is no $x \in \mathbb{N} - \{1\}$ where $0|x$. However, this is the only maximal element. To show this, suppose $a \in \mathbb{N} - \{0, 1\}$. Now $a|0$ so a is not maximal, thus 0 is the only maximal element.

3. To prove this, we have to prove both implications (directions)
   ($\Longrightarrow$) Suppose $a \in A$ is the minimum of A under $\leq$. This means that $a \leq x$ for all $x \in A$. Now $a \leq x \iff x \leq^{-1} a$, so $x \leq^{-1} a$, $\forall x \in A$. This is precisely the definition for the maximum of A under $\leq^{-1}$. This means that:
   (a is the minimum of A under $\leq$) $\Longrightarrow$ (a is the maximum of A under $\leq^{-1}$)
   ($\Longleftarrow$) Suppose $a \in A$ is the maximum of A under $\leq^{-1}$. This means that $x \leq^{-1} a$ for all $x \in A$. Now $x \leq^{-1} a \iff a \leq x$, so $a \leq x$, for all $x \in A$. This is the definition for the minimum of A under $\leq$. Thus, we have that:
   (a is the maximum of A under $\leq^{-1}$) $\Longrightarrow$ (a is the minimum of A under $\leq$)
   Finally, since both directions have been proven, we can conclude that
   (a is the minimum of A under $\leq$) $\iff$ (a is the maximum of A under $\leq^{-1}$)